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Full-stack developer

A full-stack developer is a software engineer who works on both front-end (What the users see and interact with) and the back-end (the servers, databases, and business logic that power the app) of a web application. Day to day, the role involves building and maintaining user-facing features in HTML, CSS, and JavaScript. Designing and querying relational or no SQL databases and creating REST APIs and GraphQL endpoints. Full-stack developers also collaborate with the UX designers and product managers, and review teammates' code when needed. Full-stack developers also handle concerns like containerizing apps with Docker, Running CI/CD pipelines in Github actions or Jenkins, and monitoring production services in the cloud.

A job posted in Indeed for a full-stack developer states that the core responsibilities are: Building enterprise web applications across front-end and back-end layers, developing features end-to-end using .NET, REST APIs, Angular, and SQL server, and collaborating in an Agile team with product, QA, and architecture on design and delivery. This all aligns closely with the industry descriptions that define the role as handling the full software development life cycle, from conception to deployment (InterviewGuy, 2025)

Required Skills and Tools

Front-end: HTML5, CSS3, JavaScript, and a major framework such as React, Angular, or Vue.js.

Back-end: Node.js/Express, Python/Django, Java/Spring boot, or C#.NET Core.

Databases: MySQL, PostgreSQL and MongoDB (NoSQL).

APIs: RESTful design, JSON, and increasingly GraphQL.

Version control: Git (github/gitlab).

DevOps basics: Docker, Jenkins, CI/CD pipelines, and at least one cloud provide (AWS or Azure).

Testing: Unit and integration testing frameworks like Jest, Junit, or Selenium.

Its connection to this C++ course, are some foundational concepts that are very important in the field such as control flow, functions, object-oriented programming arrays and pointers. We don't use C++ as much as other program languages but their type system all descend from the same OOP principles introduced in C++. What's new is the web-specific toolchain, javascripts asynchronous model and database query languages (SQL) and cloud infrastructure

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Educational Path

The education for a full-stack development comes from different places, such as courses, bootcamps and self-taught topics. A bachelor's degree in Computer Science or Software Engineering is the most common credential for most of them. Bootcampss like Fullstack academy or App academy are also useful and respected for many employers. Certifications like AWS Certified Developer or Google Professional Cloud Developer carry weight for the

DevOps/cloud aspects of full-stack work. Self-taught paths via platforms like The Odin Project, freecodecamp, or scrimba. The clearest across hiring managers is the Bachelor's degree because it opens doors to large enterprises.

Portfolio Projects

Project 1: Pokedex Web App.

This project involves building a browser-based pokedex that lets the user search for any pokemon by name or number, view its stats (Type, abilities, base stats), and browse a scrollable list of all 1000+ pokemons. All data is pulled live from the free PokeAPI (pokeapi.co) using the browser's built in `fetch()` function. Skills used include HTML, CSS and JavaScript. Specially loops, functions, arrays, and conditional logic which is what is being taught in C++. This is compelling to an employer because it proves that a student can read external documentation, consume a real public API, and render dynamic data in the browser.

Project 2: Full-stack personal list app with Authenticator

This project involves building a full-stack web application where users can register, log in and out, and manage a personal list of items such as saved recipes, a reading list, etc, that persists across sessions because it is stored server-side in a database. The front-end uses HTML, CSS and JavaScript to render the list and handle form submissions. The back-end is built with node.js and express, using REST API endpoints for creating, reading, updating and deleting items. User password will be hashed with bcrypt before being stored and authentication is handled with JWT tokens so the server can identify which user's data to return on each request. Data base using MongoDB with mongoose or Postgre SQL. This will be the most important and difficult project in the list because it requires a lot of knowledge and demonstrates mastery of the complete full-stack request cycle.

Project 3: Music Library Management App

This project involves building a full-featured music library app where authenticated users can catalog their personal music collection, organize it into playlist, search and filter by artist, genre, or release year. It will allow entries with album art and metadata fetched from the Spotify web API or the MusicBrainz API. This project is compelling to employers because it involves a non-trivial relational database schema with joins and aggregation queries. It shows the kind of end-to-end ownership from schema design to deployment.

Target companies

Amazon

Amazon hires thousands of software engineers across teams like AWS, Prime Video, Alexa, and Amazon Shopping. Their software engineering internships require knowledge of a programming language such as Java, Python, C++, or TypeScript, along with basic data structures and algorithms. Amazon is a great place to learn because interns work on real projects and gain experience with large-scale systems. The company is also known for its Leadership Principles, so interviews include both coding and behavioral questions.

Google

Google offers software engineering internships for products such as Search, YouTube, Google Maps, and Google Cloud. Interns need experience with a programming language like Java, C++, Python, or JavaScript and a strong understanding of computer science fundamentals. Google is known for its mentorship opportunities and technical culture. Interns usually work on real projects that impact users, making it one of the most competitive and rewarding internship programs.

Microsoft

Microsoft hires software engineering interns to work on products like Azure, GitHub, Teams, Xbox, and Microsoft 365. The internship requires programming experience and knowledge of data structures and algorithms. Microsoft provides interns with hands-on experience, mentorship, and the opportunity to contribute to real products. The company also offers the Explore Microsoft program, which is designed for students who are earlier in their college careers.

Internships

- **Amazon SDE Internship — Fall 2026:**
 - 1) Experience with at least one general-purpose programming language such as Java, Python, C++, C#, or TypeScript
 - 2) Experience with data structure implementation, basic algorithm development, and/or object-oriented design principles
 - 3) Enrolled in a Bachelor's degree or above in Computer Science, Computer Engineering, Data Science, or a related STEM field.
- **Ramp Software Engineer Internship (Android)**
 - 1) Experience programming in Java, Kotlin, Python, C++, or similar languages.
 - 2) Knowledge of software development fundamentals, including data structures and algorithms.
 - 3) Currently pursuing a Bachelor's degree in Computer Science, Software Engineering, or a related field.

- **JPMorgan Chase Software Engineer Program Internship:**

- 1) Knowledge of programming in languages such as Java, Python, C++, or similar object-oriented languages.
- 2) Understanding of data structures, algorithms, and software engineering principles.
- 3) Currently pursuing a Bachelor's or Master's degree in Computer Science, Computer Engineering, or a related technical field.

Career Outlook

Software development is one of the fastest-growing careers today. The Bureau of Labor Statistics predicts that software developer jobs will grow by **17% from 2024 to 2034**, which is much faster than the average growth for all jobs.

Full-stack developers are in high demand because they can work on both the front end and back end of websites and applications. Most full-stack developers earn good salaries, with many making between **\$105,000 and \$160,000 per year**. Entry-level jobs often start between **\$80,000 and \$110,000**.

Another benefit is flexibility. Many companies offer remote or hybrid work, so developers can work from home or from different locations while still earning competitive pay.

The demand for full-stack developers stays high because almost every company needs websites or web applications. According to a 2025 developer survey, full-stack development was the most common job role among developers worldwide.

Personal Reflection

I chose this field because I enjoy learning about the front-end and back-end environment as it is today, maybe I don't know much about it in depth but I know that I will one day. I like the design and logic behind the programming and I know it will be a challenge to learn everything

and make something that work correctly on the inside and feels right on the outside. My head starts come from this C++ course, because it is teaching me to organize code into functions and classes, ho to think about data in arrays and structures, and how to read from and write to files. My biggest gap is the web layer, because I have never used or worked with JavaScript, HTTP, CSS, or any other database. It will take me a well amount of practice and reading/learning to master it. But I feel like one concrete thing I could do is start learning all of them and maybe build one of those projects mentioned in this document. I feel like I could try to build the Pokedex project because it only requires HTML, CSS and JavaScripts, no frameworks, no servers, and all the logic of fetching data from an API, looping through arrays, and displaying them maps onto the loops, arrays and conditional that we are practicing in this C++ class right now. I could finish it and push it into my github as a public repository. I believe it would be a great challenge a meaningful first step towards my goal of becoming a full stack developer.

Work Cited

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Research Sources Used:

Job Postings: Indeed, Amazon Jobs, JPMorgan Chase Careers, Ramp Careers

Salary and Market Data: U.S. Bureau of Labor Statistics Occupational Outlook Handbook, Glassdoor

Company Research: Amazon Science & Engineering Blog, JPMorgan Chase Technology Blog, Ramp Engineering Blog, Glassdoor Company Reviews

Project Inspiration: GitHub Trending Repositories, Reddit

Industry Trends: Stack Overflow Developer Survey: <https://survey.stackoverflow.co/2025/>